

A proof of the conjectured recurrence for OEIS A127905

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Abstract

OEIS A127905 carries a conjectured linear recurrence with polynomial coefficients, of order 3. The entry gives no generating function, but it does give the sequence as a single hypergeometric sum. We prove the conjecture without assuming any generating function: creative telescoping produces, from the summand alone, an operator of order 3 that annihilates the sequence, together with a rational certificate that is checked as an exact identity rather than trusted. The conjectured operator is then shown to be a left multiple of it by division in the Ore algebra $\mathbb{Q}(n)[N]$. That last step is what the conjecture needs: the two operators have the same order but are not proportional, so the conjecture still has to be derived from what telescoping gives.

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1 The sequence and the conjecture

OEIS A127905 is “Construct triangle in which n -th row is obtained by expanding $(1+x+x^3)^n$ and take the next-to-central column”. It has offset 0 and begins

$$a(0), \dots, a(7) = 0, 1, 2, 3, 8, 25, 66, 168, \dots$$

The entry states, not as a conjecture,

$$a(n) = \text{Sum}_{k=0.. \lfloor (n-1)/3 \rfloor} (3^{*k+1}) * C(n, 3^{*k+1}) * C(3^{*k}, k) / (2k+1).$$

so that, writing $F(n, k)$ for the summand,

$$a(n) = \sum_{k=0}^{\lfloor \frac{n-1}{3} \rfloor} F(n, k), \quad F(n, k) = \frac{(3k+1) \binom{3k}{k} \binom{n}{3k+1}}{2k+1}. \quad (1)$$

This is the only input the proof takes from the entry, and it was checked against the entry’s own DATA before use.

The entry separately carries the following comment:

Conjecture: $2^{*}(2^{*}n+1)^{*}(n-1)^2^{*}a(n) - 2^{*}n^{*}(6^{*}n^2 - 12^{*}n + 5)^{*}a(n-1) + 6^{*}n^{*}(n-1)^{*}(2^{*}n-3)^{*}a(n-2) - 31^{*}n^{*}(n-1)^{*}(n-2)^{*}a(n-3) = 0$. - R. J. Mathar, Feb 23 2015

As of the “Last modified” line on the live entry (2022-09-08, revision 17) the statement is still recorded as a conjecture, and no proof appears among the entry’s links, formulas or programs.

2 Telescoping the summand

Let N denote the shift in n , $(Nf)(n) = f(n+1)$, and Δ_k the forward difference in k , $(\Delta_k g)(k) = g(k+1) - g(k)$.

Lemma 1. *Suppose there are rational functions $\sigma_0, \dots, \sigma_r$ of n , not all zero, and a rational function $R(n, k)$ such that*

$$\sum_{i=0}^r \sigma_i(n) F(n+i, k) = \Delta_k [R(n, k) F(n, k)]. \quad (2)$$

Suppose further that $F(n, k) = 0$ for every k outside the range of summation in (1). Then $\sum_i \sigma_i(n) a(n+i) = 0$.

Proof. Because $F(n, k)$ vanishes off the stated range, the sum in (1) may be taken over all $k \in \mathbb{Z}$ without changing it, and the same holds for each $F(n+i, k)$. Summing (2) over all k , the left side becomes $\sum_i \sigma_i(n) a(n+i)$ and the right side telescopes: the partial sums of a forward difference collapse to the boundary values, which are 0 because $R(n, k) F(n, k)$ vanishes for $|k|$ large. Hence the left side is 0. $\square$

The point of Lemma 1 is that (2) is an identity between rational functions once both sides are divided by $F(n, k)$: the quotients $F(n+i, k)/F(n, k)$ and $F(n, k+1)/F(n, k)$ are rational because F is hypergeometric in both variables. So a proposed (σ, R) can be *checked*, exactly, whatever produced it.

For A127905 such a pair exists with $r = 3$:

$$\sigma_0(n) = -\frac{31}{108}, \quad \sigma_1(n) = \frac{2n+3}{18n+18}, \quad \sigma_2(n) = \frac{-6n^2-24n-23}{54n^2+162n+108}, \quad \sigma_3(n) = \frac{2n^2+11n+14}{54n^2+216n+162}$$

and certificate

$$R(n, k) = \frac{2k^3 + k^2}{54k^3 - 54k^2n - 54k^2 + 18kn^2 + 36kn + 12k - 2n^3 - 6n^2 - 4n}.$$

Dividing (2) by $F(n, k)$ and cancelling gives 0 identically, so Lemma 1 applies and

$$L = \sum_{i=0}^3 \sigma_i(n) N^i \quad (3)$$

annihilates a .

3 From the derived recurrence to the conjectured one

The operator (3) is not the conjectured one. It has the same order as the conjectured one but is not a multiple of it by a constant. The two are connected by division in the Ore algebra $\mathcal{A} = \mathbb{Q}(n)[N]$, where multiplication obeys $Nq(n) = q(n+1)N$.

Lemma 2. *If $C = QL$ in $\mathcal{A}$ and $L(a) = 0$, then $C(a) = 0$ at every n where the coefficients of Q are defined.*

Proof. $C(a) = (QL)(a) = Q(L(a)) = Q(0) = 0$. The coefficients of Q are rational functions of n , so the step is valid away from their poles. $\square$

Writing the conjectured recurrence as an operator C and dividing by L leaves remainder 0, with quotient

$$Q = (108n^3 + 648n^2 + 1188n + 648).$$

No coefficient of Q has an integer pole, so Lemma 2 applies at every index.

Theorem 3. *The conjectured recurrence*

$$2(n-1)^2(2n+1)a(n) - 2n(6n^2 - 12n + 5)a(n-1) + 6n(n-1)(2n-3)a(n-2) - 31n(n-2)(n-1)a(n-3)$$

holds for every $n \geq 3$.

Proof. By Lemma 1 the operator (3) annihilates a , and by Lemma 2 applied to $C = QL$ so does the conjectured operator. No coefficient of Q has an integer pole. $\square$

4 Verification

Every number above is an output of a script written separately from this note, and four independent checks were run.

First, the certificate was verified, not trusted: (2) was divided by $F(n, k)$ and the difference of the two sides cancelled to 0 as a rational function of n and k .

Second, the vanishing of $F(n, k)$ off the range of summation — the hypothesis Lemma 1 needs, and the one that is easy to assume without looking — was checked directly, on concrete integers on both sides of the range.

Third, the factorisation was confirmed by multiplying back: QL was expanded in $\mathcal{A}$ and its coefficients cancelled against those of C to zero.

Fourth, the conjecture itself was evaluated on the published terms, in exact integer arithmetic and without reference to any operator: for each n in range the sum $\sum_i p_i(n)a(n-i)$ was formed from the entry's own values and found to vanish, for all 24 values of n from $n = 3$ upward. The formula (1) was likewise checked term by term against the DATA section before being used.

References

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